

Project Title

- AutoFixer: AI Agent for Automatic Bug Detection & Program Repair

Problem Statement

- Software bugs cause failures.
- Goal: Build AI that detects bugs and generates correct patches automatically.

Motivation

- Traditional debugging is slow.
- AI-based automated program repair is cutting-edge and rare.

Objectives

- - Detect bugs in source code
- - Generate multiple patch candidates
- - Validate patches using unit tests
- - Output best correct patch

Architecture

- 1. Input Processing
- 2. Patch Generation (CodeT5)
- 3. Patch Application
- 4. Test Execution
- 5. Patch Verification
- 6. Output Result

Model Used

- CodeT5 / Transformer Seq2Seq
- Fine-tuned on buggy → fixed code pairs

Dataset

- - CodeXGLUE
- - Defects4J
- - QuixBugs
- - Synthetic generated bugs

Methodology

- 1. Prepare dataset
- 2. Train model
- 3. Implement patch generator
- 4. Implement automated testing loop
- 5. Evaluate functional repair rate

Evaluation Metrics

- - Functional Repair Rate
- - BLEU/Exact Match
- - Compilation Success Rate
- - Patch Validity

Demo Workflow

- Run tests → Detect failure → AI suggests patch → Apply patch → Re-test → Output fixed code

Results

- Expected: 30-40% functional repair rate on simple bugs

Future Scope

- - Multi-language repair
- - AST-level patching
- - Integration with IDEs

Conclusion

- AutoFixer shows the power of AI for automated program repair.